BUILDING SIMULATION 2025 24-27 AUG BRISBANE CARBON AND CLIMATE RESPONSIVE

International Building Performance Simulation Association

Reducing the gap between predicted and measured urban microclimate through handling parameter uncertainty and model form uncertainty

Gigih R. Setyantho[1], Yeonsook Heo[1*]

[1]School of Civil, Environmental, and Architectural Engineering, Korea University, Seongbuk-gu, Seoul, Republic of Korea

*Corresponding Author: yeonsookheo@korea.ac.kr

Abstract

Physics-based models, particularly urban canopy models (UCM), have been widely used to investigate urban microclimate conditions of various urban areas. Despite its developments, deployment of UCM for city-scale urban simulations presents major challenges as follows: (1) UCM includes many uncertain input parameters, referred to as parameter uncertainty, due to the lack of data availability; (2) UCM is based on the concept of urban canyon that assumes homogeneous urban settings with one representative building geometry, which may yield model bias. This study aims to improve wind speed predictions in the vertical city weather generator (VCWG), one of the UCMs, through calibration of key parameters and investigation of intermediate model variables to identify potential sources of model bias. 71 weather stations are deployed in different vertical layers: urban canopy-level and roughness sublayer-level, which allows for improving UCM prediction in different urban characteristics and vertical heights. Results demonstrate that parameter calibration reduced the RMSE of wind speed prediction from 1.81 m/s to 1.78 m/s. Additionally, this study provides insights into model discrepancies related to turbulence and pressure gradient formulations, as the foundation for further improvements in urban wind modeling.

Key Innovations

- Improving the prediction accuracy of UCM by calibrating key uncertain input parameters, such as aerodynamic roughness length and turbulent diffusion coefficients.
- Identifying potential areas for model improvements, specifically about model-form uncertainty.

Practical Implications

This study provides evidence to correctly set up physical parameters that are typically assumed in the UCM and presents the calibration of UCMs on the basis of actual measurement data under actual urban characteristics.

Introduction

Urbanization has significantly increased the global urban population, projected to rise from 55% in 2019 to 68% by 2050 (United Nations Department of Economic and Social Affairs, 2019). As urbanization expands, increased infrastructure and population density will further alter local climate conditions (Oke, 2002). As one of the climate variables, urban wind plays a key role in local climates, notably affecting urban heat island (UHI) intensity and air pollution levels. Studies have shown that a 1 m/s increase in wind speed can reduce UHI intensity by 0.14°C in Melbourne, Australia (Morris & Simmonds, 2001) and lower PM2.5 concentrations by 0.16 µg/m³ in China (Liu et al., 2020). However, urban wind patterns are highly variable due to the complex interactions within the urban boundary layer (UBL), particularly in the canopy and roughness sublayers, where wind flow is strongly influenced by building height, street layouts, and other urban morphological features (Ricciardelli & Polimeno, 2006).

To predict urban wind conditions, the computational fluid dynamics (CFD) approach is commonly used due to a detailed full three-dimensional analysis of the entire calculation domain (Blocken, 2015). CFD models use different types of turbulence algorithms, such as Reynolds-averaged Navier-Stokes (RANS) and large eddy simulation (LES), in which LES offers higher precision compared to RANS (Wang et al., 2017). However, their high computational cost limits their application in City-scale or longer-duration urban studies (Blocken, 2015).

Another approach for modeling microclimate conditions is the use of urban canopy models (UCM), which simplify urban characteristics into a canyon concept to represent the interaction between urban surfaces, vegetation, and atmospheric conditions. Several UCMs have been developed, initiated by the Town Energy Balance (TEB) model (Masson, 2000). Later, the urban weather generator (UWG) was developed by coupling TEB with building energy simulation to predict microclimate variables, primarily focusing on canyon temperature prediction (Bueno et al., 2013). More recently, the vertical city weather generator (VCWG) enhanced UCM models with vertical profile predictions on microclimate variables, including temperature, wind speed, and specific humidity (Moradi, 2021).

The validation of UCMs had been done in previous studies. Commonly, the validation was against the measurement data in Basel, Switzerland (Bueno et al., 2013; Moradi et al., 2021). Additionally, the VCWG had been validated against measurement data in Vancouver, Canada (Moradi, 2021), which demonstrated reasonably accurate predictions of temperature and humidity

https://doi.org/10.26868/25222708.2025.1459

variables. However, these validations revealed that wind speed predictions had lower accuracy compared to other microclimate variables (Moradi, 2021).

Although the development of UCMs has advanced, several challenges remain when applying UCMs for city-scale prediction. First, previous studies identified that UCM simulations require numerous uncertain input parameters, often due to limited information availability (Mao et al., 2017; Setyantho et al., 2024). To address this, previous studies identified and calibrated important parameters for temperature prediction in UWG (Mao et al., 2017) and for wind speed prediction in VCWG (Setyantho et al., 2024), which led to reduced prediction gaps. Second, a key assumption in UCMs is the simplification of urban areas into street canyons with uniform building geometries (Chen et al., 2011). A previous study used a 250-meter buffer (Rotach et al., 2005) to define the urban characteristics in Basel, which was later used to validate UWG (Bueno et al., 2013) and VCWG (Moradi, 2021). Meanwhile, Mao et al. calculated the urban variables based on the average of rectangular-shaped districts. These studies highlight the role of parameter uncertainty in model accuracy performance. Moreover, a limited number of stations used for model validation may be insufficient to examine the UCMs and assess the potential model bias due to different urban characteristics.

This study aims to reduce the gap between predicted and measured urban wind speeds through the calibration of important parameters, such as aerodynamic roughness length and turbulent diffusion coefficients. It also extends the previous work by investigating the intermediate model variables to better understand potential sources of prediction bias (Setyantho et al., 2024). First, we applied VCWG v1.4.5 (Aliabadi et al., 2021) to predict wind speed at 71 weather stations. Second, we calibrated significant parameters and assessed their impact on prediction accuracy across all stations. Third, we analyzed how urban morphology influences intermediate model variables to identify potential sources of model bias. Finally, we discussed potential areas for VCWG model improvement for wind speed prediction.

Methodology

The research framework for this study is depicted in Figure 1. Wind speed data from multiple weather stations were used to calibrate the multi-layer urban canopy model (MLUCM), based on the vertical city weather generator (VCWG) version 1.4.5. The process began with the extraction of urban variables from geographical information system (GIS) datasets, followed by the development of the baseline VCWGv1.4.5 model. Next, model calibration was performed to reduce the discrepancy between measurements and predictions. Model calibration focused on significant parameters identified by local sensitivity analysis in previous studies (Setyantho et al., 2024). The calibration used a Latin hypercube sampling (LHS) approach to explore parameter combinations. Furthermore, we investigated the potential model bias based on urban characteristics in different locations.

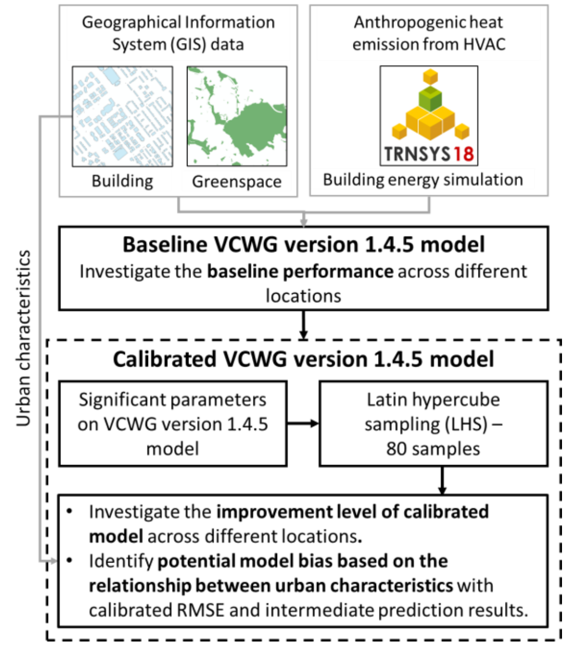

Figure 1. Research framework flowchart

Vertical city weather generator (VCWG) version 1.4.5 model

In this study, we used a specific urban canopy model, called the VCWG model version 1.4.5 (Moradi, 2021). This model was proposed to improve the previous urban canopy model, the urban weather generator (UWG) (Bueno et al., 2013). The VCWG model introduced key improvements compared to previous urban canopy models, including the ability to predict temperature, wind speed, and humidity at various heights within the urban boundary layer. Notably, wind speed prediction, which was absent in UWG, was incorporated into VCWG. Figure 2 illustrates a simplified schematic of the VCWG model, which is coupled with multiple sub-models. The Monin-Obukhov Similarity Theory sub-model accounted for thermal stability and roughness length to generate the vertical profiles at rural weather stations. Additionally, the vertical diffusion sub-model resolved the weather variables in urban areas at various heights (Aliabadi et al., 2021; Moradi et al., 2021).

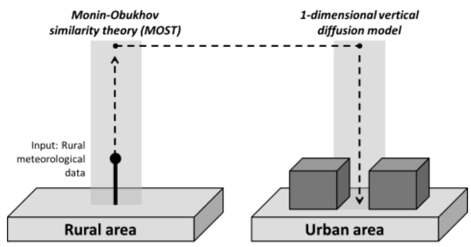

Figure 2. Simplified schematic of the VCWG model

The VCWG model required four categories of input parameters: urban, vegetation-related, rural weather stations, and physical constants. While vegetation-related parameters and physical constants were straightforward, urban and rural parameters were required to characterize the urban morphology and thermal properties of the

https://doi.org/10.26868/25222708.2025.1459

respective locations. For wind speed prediction, VCWG used two key urban morphology variables, such as building coverage ratio (BCR) to represent urban density and frontal area index (FAI) to represent the projected building surface area to the wind direction. More detailed methods and calculations to derive VCWG input variables in urban locations were described in the previous study (Setyantho et al., 2024).

Urban wind speed prediction in the VCWG model was calculated based on the Reynolds-averaged equations for the 1-dimensional vertical diffusion model. Equation (1) represents the momentum equations for the x- (U) and y- (V) components, which govern wind speed prediction. This equation incorporates three key terms: turbulent flux of momentum (Term I), acceleration due to pressure gradient (Term II), and the sum of drag pressure from the building and vegetation (Term III). More detailed mathematical formulations and parameterizations for the vertical profile of temperature, turbulence kinetic energy, and humidity were provided by Moradi et al. (2021), and the supporting documentation of VCWG v1.4.5 (link: https://github.com/AmirAAliabadi/VCWGv1.4.5/tree/main).

$$\begin{aligned} \frac{\partial \bar{U}}{\partial t} &= \underbrace{\frac{\partial \overline{uw}}{\partial z}}_{[Term\ I]} - \underbrace{\frac{1}{\rho}\frac{\partial \bar{P}}{\partial}}_{[Term\ II]} - \underbrace{D_x}_{[Term\ III]} \\ \frac{\partial \bar{V}}{\partial t} &= \underbrace{\frac{\partial \overline{vw}}{\partial z}}_{[Term\ I]} - \underbrace{\frac{1}{\rho}\frac{\partial \bar{P}}{\partial y}}_{[Term\ II]} - \underbrace{D_y}_{[Term\ III]} \end{aligned} \tag{1}$$

Previous studies identified three significant input parameters based on local sensitivity analysis in the VCWG model that influence wind speed prediction by affecting the momentum equation. Those parameters were the turbulent diffusion coefficient during unstable conditions ($C_{k,unstable}$), pressure gradient coefficient (C_{dpdx}), and aerodynamic roughness length ($R_{z0,aero}$) (Setyantho et al., 2024). Among these parameters, $C_{k,unstable}$ indicated a more significant role in wind speed prediction compared to others. Equation (2) describes how $C_{k,unstable}$ is used to compute the diffusion coefficient K_m, which governs the turbulent momentum flux (Term I in Equation (1)). Similarly, K_m is calculated using $C_{k,stable}$ under stable conditions, as shown in Equation (2), depending on the thermal stability conditions applied at each time step.

$$\begin{aligned} K_m &= C_{k,unstable} \times l_k \times \sqrt{TKE} \\ K_m &= C_{k,stable} \times l_k \times \sqrt{TKE} \end{aligned} \tag{2}$$

where l_k is a length scale derived from CFD-based sensitivity analysis (Nazarian et al., 2020). The TKE is computed using a prognostic equation that accounts for shear production, turbulent transport, buoyancy effects, wake generation from urban elements, and dissipation (Krayenhoff et al., 2015). Buoyant production reflects vertical thermal gradients, thereby linking temperature structure directly to momentum diffusion. Vertical temperature profiles are computed using an energy transport equation that includes turbulent heat diffusion and multiple source and sink terms: roof, ground, and wall heat fluxes, urban vegetation effects, radiative divergence, and waste heat from buildings (Moradi et al., 2021). In this study, the default VCWG building energy model was replaced with TRNSYS simulations of prototype typical Korean residential and non-residential buildings. Hourly cooling and heating demands were generated for residential and non-residential building types and normalized by building footprint area. Hence, these demands were used to compute total sensible waste heat, a detailed calculation is in Setyantho et al. (2024).

Meanwhile, Equation (3) shows how C_{dpdx} is used to compute the pressure gradient as in Term II in Equation (1). This term is also applied as a boundary condition in the one-dimensional vertical diffusion model, which predicts wind speed within urban areas. The friction velocity (u_*) in Equation (3), is determined based on the MOST sub-model (Equation (4)), where it is a function of $R_{z0,aero}$ and rural wind speed (U_{ref}). These equations captured the interaction between turbulent diffusion, pressure gradient, and surface roughness effects, which form the basis for wind speed predictions in VCWG.

$$\begin{aligned} \frac{\partial \bar{P}}{\partial x} &= C_{dpdx} \times \frac{\rho u_*^2 \cos\theta_s}{H_{top}} \\ \frac{\partial \bar{P}}{\partial y} &= C_{dpdx} \times \frac{\rho u_*^2 \sin\theta_s}{H_{top}} \end{aligned} \tag{3}$$

$$u_* = f(R_{z0,aero}, U_{ref}, \kappa, R_{disp}) \tag{4}$$

Model calibration

In this study, we calibrated the significant parameters identified by local sensitivity analysis, including turbulent diffusion coefficients during unstable conditions ($C_{k,unstable}$) and aerodynamic roughness length ($R_{z0,aero}$). Although $C_{k,unstable}$ indicated more significant as the higher sensitivity index, the turbulent diffusion coefficient during stable conditions ($C_{k,stable}$) also showed a non-negligible sensitivity index and was included in the calibration. The C_{dpdx} was set to a constant value of 1.0 (same as the baseline value of VCWG) and was not calibrated to preserve a constant pressure gradient.

We used the Latin hypercube sampling algorithm to obtain 80 parameter sets by varying each significant parameter value. This approach minimized the number of required simulations due to high computational time required for multiple locations in the VCWG model. The LHS approach ensured even coverage compared to the random sampling method (McKay et al., 2000). We used an open-source Python library called SALib (Herman & Usher, 2017) to draw the sample from LHS. We calculated the overall RMSE ($RMSE_{all}$ in Equation (5)) to evaluate each of the 80 parameter combinations generated by the LHS approach. Equation (5) consisted of the simulated values (s_i), the measured values (m_i), and the total number of observations across all stations and days (N). Additionally, we calculated station-specific RMSE ($RMSE_{station}$ in Equation (6)), where n is the number of observations for a single station across all days. By

https://doi.org/10.26868/25222708.2025.1459

calculating both $RMSE_{all}$ and $RMSE_{station}$, we assessed VCWG's overall model performance and its accuracy at individual stations, respectively.

$$RMSE_{all} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(s_i - m_i)^2} \quad (5)$$

$$RMSE_{station} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(s_i - m_i)^2} \quad (6)$$

Case description

This study was conducted in Seoul, the capital city of the Republic of Korea, which covers an area of approximately 605.25 km^2. A total of 71 weather stations were used to capture local climate conditions across the city, as illustrated in Figure 3, which ensured comprehensive coverage across diverse urban environments within the metropolitan area. These stations were obtained from two datasets. The first dataset was 232 SK Telecom weather stations (SK stations), which had been used in previous studies (Hong & Heo, 2021). The second dataset was Seoul city weather stations (S-DoT stations), which had also been used in previous studies (Hong & Heo, 2023). Although the S-DoT weather stations include 1,019 sensor units, only a subset is equipped with wind speed sensors. Therefore, we selected 54 SK stations and 17 S-DoT stations that meet the VCWG urban criteria range and are located within the urban canopy layer and roughness sublayer. In this study, the urban canopy layer is defined as the height from the ground up to one time the average building height (1×BH), while the roughness sublayer is defined as ranging from one to two times the average building height (1×BH to 2×BH) (Barlow, 2014). The average building height was calculated within a 250-meter buffer around each weather station.

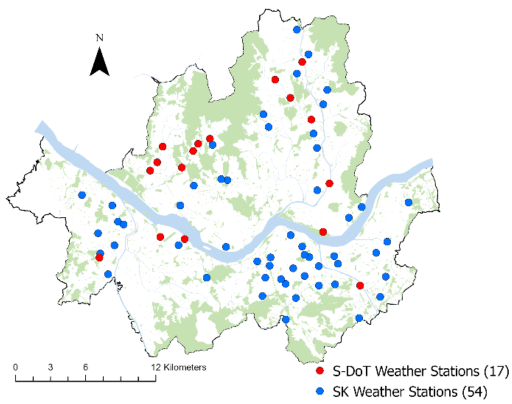

Figure 3. Weather stations map in Seoul

One of the initial steps to set up the VCWG model was the location of a rural weather station as the reference meteorological condition, specifically one with surrounding obstacles less than 10 meters. In this study, the automated surface observing system (ASOS) station in Icheon was chosen as the rural weather station, as shown in Figure 4. This station is located approximately 50 km southeast of central Seoul and managed by the Korea Meteorological Administration (KMA). The ASOS at Icheon is situated in an open agricultural area with low surrounding obstacles, making it well-suited for the VCWG rural station criteria.

Figure 4. Location of the rural weather station in Icheon

Another important step in setting up the VCWG model was the calculation of urban variables as the input variables. This study used a 250 m radius buffer to define the extent of urban morphology calculations. Figure 5 presents the distribution of key urban variables, including building height, building coverage ratio, frontal area index, and sensor height across different local weather station datasets. The urban variables for SK and S-DoT stations showed similar distributions, except for sensor height, which exhibits a notable difference. S-DoT sensors were positioned between 2.4 – 5.5 m to capture pedestrian-level conditions, while SK stations range from 3.5 – 62.5 m, often on rooftops or other elevated locations. These elevated sensor positions may be influenced by local wind acceleration, wake effects, or thermal biases due to proximity to building materials and surfaces. While such placements may not fully represent the ambient wind field, SK and S-DoT weather stations are valuable to evaluate VCWG performance across multiple urban layers, including the canopy layer and roughness sublayer.

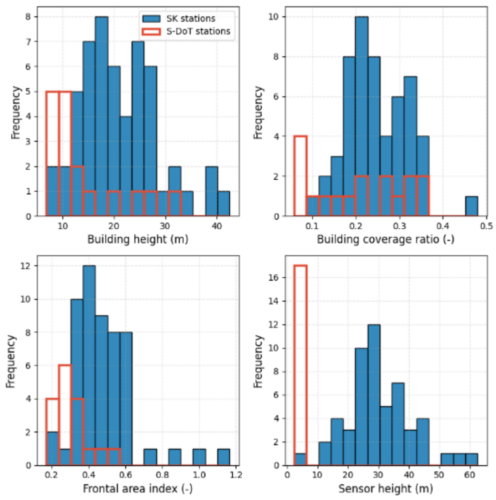

Figure 5. Distribution of key urban characteristics used in VCWG

Our local weather station datasets covered different time periods. SK stations provided data from 2017, while S-DoT data was fully available starting from 2021. The KMA dataset, used for rural weather stations, had continuous records from 1972. To ensure consistency, we

https://doi.org/10.26868/25222708.2025.1459

selected four non-precipitation days based on the rural weather station. Table 1 summarizes the weather conditions for the selected days in 2017 and 2021, to represent summer, winter, and transitional seasons, to maintain comparability between the SK and S-DoT model evaluations. Additionally, a one-day spin-up period was applied before each selected day to ensure the stability of VCWG outputs.

Table 1. Selected days for the model calibration

	Date (YYYY/MM/DD)	Average daily temperature (℃)	Average daily wind speed (m/s)
SK stations	2017/05/29	22.4	1.47
	2017/12/07	0.4	1.32
	2017/04/04	12.3	1.39
	2017/02/18	-1.7	2.28
S-DoT stations	2021/06/22	21.9	0.89
	2021/01/17	-5.6	1.33
	2021/04/10	11.5	1.27
	2021/11/23	0.2	1.82

Results

Baseline model performance

This section evaluated the performance of the baseline VCWGv1.4.5 model. Figure 6 presents the histogram of $RMSE_{station}$, which shows a broad range of accuracy from 0.37 to 6.89 m/s and an overall $RMSE_{all}$ of 2.15 m/s. The significant variability of $RMSE_{station}$ suggested that the baseline model struggles to accurately capture local wind conditions in certain locations.

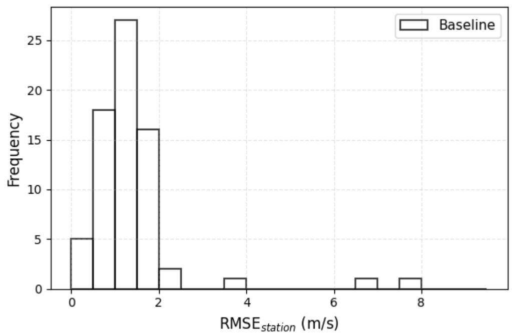

Figure 6. Histogram of RMSE at each station for the baseline model

Calibrated model performance

This section presents the calibration results for the VCWG v1.4.5 model. Table 2 summarizes the baseline and range values for significant parameters. We defined the range of aerodynamic roughness length ($R_{z0,aero}$) values based on the constraint in the VCWG model. The turbulent diffusion coefficients ($C_{k,unstable}$, $C_{k,stable}$) was calibrated by sampling a range of values ±100% of the baseline (0.4) due to a lack of reference values.

Table 2. Calibrated parameter values

Parameters & accuracy metric	Baseline	Range values	Calibrated
$R_{z0,aero}$	0.125	0.05 – 0.5	0.084
$C_{k,unstable}$	0.4	0 – 0.8	0.564
$C_{k,stable}$	0.4	0 – 0.8	0.265
$RMSE_{all}$	1.81 m/s	-	1.78 m/s

Figure 7 presents the histogram of $RMSE_{all}$ for all locations across the 80 LHS samples and the distribution of $R_{z0,aero}$, $C_{k,unstable}$, and $C_{k,stable}$ for the five parameter sets with the lowest $RMSE_{all}$ values. The baseline $RMSE_{all}$ model was 1.81 m/s (black dashed line in Figure 6(top left)), while the five lowest $RMSE_{all}$ values have lower $RMSE_{all}$ compared to the baseline model. The lowest $RMSE_{all}$ in LHS samples was 1.78 m/s, which corresponds to a calibrated $R_{z0,aero}$ value of 0.084, $C_{k,unstable}$ value of 0.564, and $C_{k,stable}$ value of 0.265. The result shows that a lower value of $R_{z0,aero}$ compared to the baseline (0.125) was selected in Figure 6(bottom left). Meanwhile, a higher value of $C_{k,unstable}$ shown in Figure 7(top right), compared to the baseline value (0.4), may suggest that the baseline model underestimated turbulent diffusion under unstable conditions. In contrast, the distribution of $C_{k,stable}$ (Figure 7(bottom right)) appeared more uniform, which indicates less adjustment is required for stable conditions.

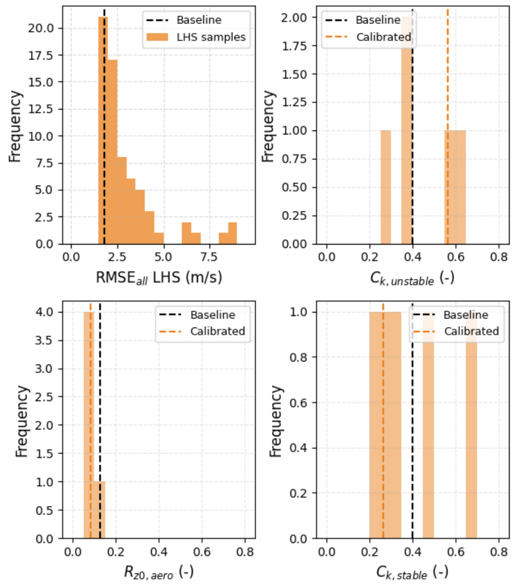

Figure 7. Histogram of RMSE for all locations (80 LHS samples) (top left) and calibrated values for the 5 lowest RMSE samples

Based on the lowest RMSE from the LHS combinations, Figure 8 illustrates the RMSE difference between baseline and calibrated models across 71 weather stations. The improvement level (IL) quantified the difference between $RMSE_{station}$ for the baseline and calibrated models. Calibration led to improvements of up to 0.2 m/s at certain locations. However, a few locations indicated slight degradation up to 0.03 m/s from baseline model

https://doi.org/10.26868/25222708.2025.1459

performance. Still, the average of IL across the 71 stations achieved 0.03 m/s. This might suggest that while parameter calibration enhanced wind speed predictions overall, some locations may require additional considerations, such as further refinement of urban morphology representation or addressing model-form uncertainties.

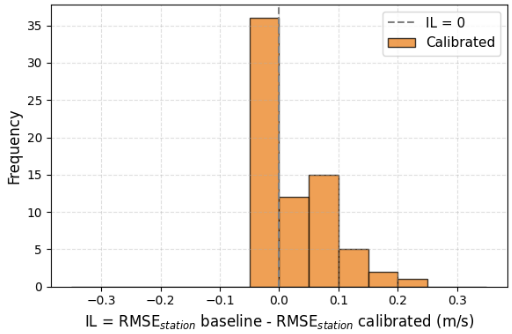

Figure 8. Histogram of improvement level (IL) of the calibration model at each station

To assess potential sources of model discrepancies, we examined the relationship between urban characteristics, intermediate variables from the VCWG model, and calibrated RMSE$_{station}$. The urban characteristics considered in this analysis include building height, building coverage ratio, frontal area index, and sensor height. While RMSE$_{station}$ reflects model performance, the choice of intermediate variables requires consideration of the calibrated values and their role in wind speed prediction. In Equation (1), two key terms depend on significant input parameters: turbulent flux of momentum (Term I) and acceleration due to pressure gradient (Term II). Since $C_{k,unstable}$ was identified as a significant parameter and was closely linked to urban characteristics, this study focused on analyzing the turbulent flux of momentum and its associated intermediate variable, turbulent diffusion (K_m).

Figure 9 shows the scatter plots between K_m (top row) and RMSE$_{station}$ (bottom row) against urban characteristics. Since K_m varied across location and time, we represented the temporally averaged values with circles and used error bars to indicate temporal standard deviations. There was no dominant trend identified between K_m and RMSE with building height, building coverage ratio, and frontal area index. However, the result in Figure 9(d) showed that the turbulent diffusion K_m increased with sensor height, which suggested stronger turbulence at higher elevations. This may illustrate that significant variables influencing wind speed prediction, such as K_m, are closely linked to the urban context. It indicates that potential model bias could be derived from spatial variations in sensor height and local urban conditions.

Discussion

This study aimed to improve urban wind speed prediction by reducing discrepancies between the VCWG model and observational data. Calibration focused on three key input parameters: the turbulent diffusion coefficient during unstable conditions ($C_{k,unstable}$), the turbulent diffusion coefficient during stable conditions ($C_{k,stable}$), and the aerodynamic roughness length ($R_{z0,aero}$). The results demonstrated that higher $C_{k,unstable}$, lower $C_{k,stable}$, and lower $R_{z0,aero}$ values improved model performance. While this study primarily addressed input parameter uncertainty through calibration, the remaining prediction gap suggested that other sources of uncertainty, specifically model-form uncertainty, may also play a role. The VCWG model relied on multiple sub-models, many of which were based on experimental or CFD simulation datasets that may not fully represent real-world conditions

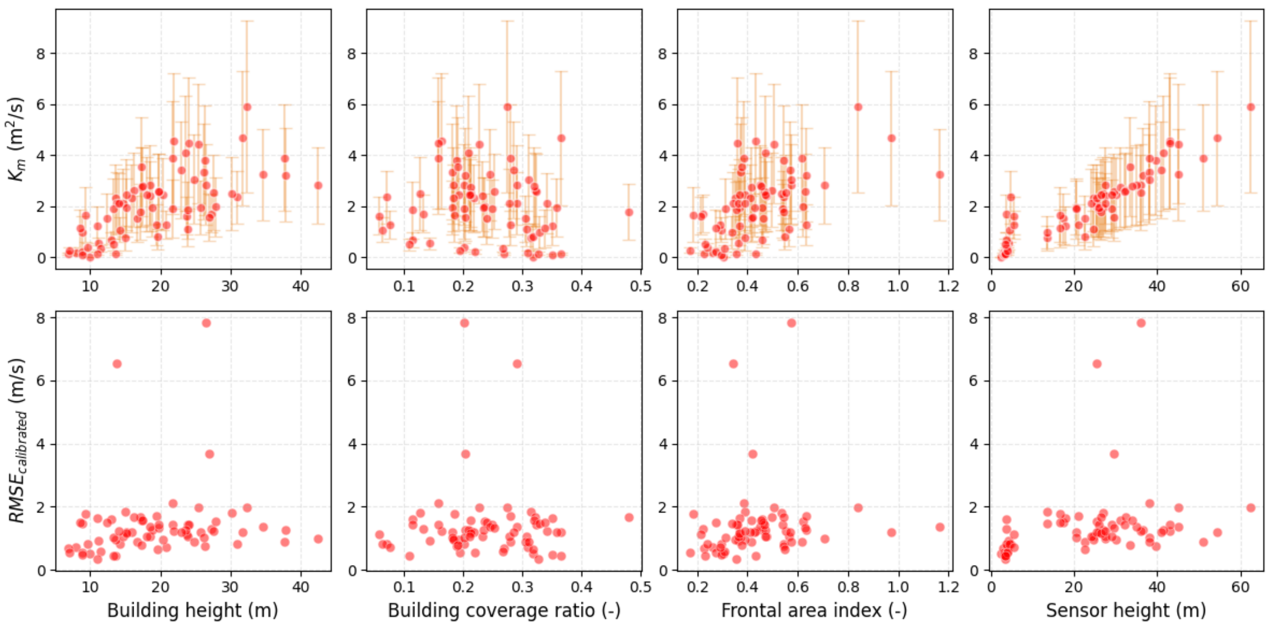

Figure 9. Scatter plots for urban characteristics with turbulent diffusion K_m (top) and calibrated $RMSE_{station}$ (below). Turbulent diffusion K_m are temporally averaged, with error bars showing temporal standard deviation

https://doi.org/10.26868/25222708.2025.1459

(Krayenhoff et al., 2015; Moradi, 2021; Nazarian et al., 2020). This limitation in model formulation could also contribute to structural biases in wind speed predictions, which were often overlooked in urban canopy modeling.

In the calibrated results, $R_{z0,aero}$ reflects the characteristics of a single rural reference station, which serves as the boundary condition for urban wind speed prediction across all 71 weather stations. Meanwhile, $C_{k,unstable}$ and $C_{k,stable}$ are applied within the urban domain to predict wind speed across all 71 weather stations. Both $C_{k,unstable}$ and $C_{k,stable}$ directly influence the turbulence diffusion as shown in Equation 2, where the spatial variables are reflected in the length-scale (l_k) parameterization. In VCWG, l_k is parameterized with the function of $l_\varepsilon/C_\varepsilon$ and C_μ, as shown in the Equation (7) and Equation (8). These functions rely on key parameters of α_1, α_2, d, and C_μ to define length scales across the vertical height. Specifically, VCWG divided the urban layer into: (a) inside the canopy ($z/BH < 1$), (b) well above the canopy ($z/BH > 1.5$), and (c) zone of transition ($1.0 < z/BH > 1.5$).

$$\frac{l_\varepsilon}{C_\varepsilon} = \begin{cases} \alpha_1(BH - d),\ for\, \dfrac{z}{BH} < 1 \\ \alpha_1(z - d),\ for\ 1 < \dfrac{z}{BH} < 1.5 \\ \alpha_2(z - d_2),\ for\, \dfrac{z}{BH} > 1.5 \end{cases} \tag{7}$$

$$\frac{l_k}{C_k} = C_\mu \frac{l_\varepsilon}{C_\varepsilon} \tag{8}$$

The VCWG model primarily adopted the parameterization scheme from Nazarian et al. to define α_1, α_2, d, and C_μ, which is derived from the LES CFD model. However, previous studies proposed different approaches to define α_1, α_2, d, and C_μ, as summarized in Table 3.

Table 3. Comparison of parameters in length-scale parameterization

Reference	α_1	α_2	d	C_μ
Santiago & Martilli, 2010	2.24	1.12	0.86	0.09
Krayenhoff et al., 2015	1.95	1.07	$BCR^{0.15}$	0.09
Nazarian et al., 2020	4	$f(BCR)$	$BCR^{0.15}$	$f(BCR, z/BH)$
Lu et al., 2024	5.5	$f(BCR)$	$BCR^{0.15}$	$f(BCR, z/BH)$

For instance, Santiago and Martilli applied constant values to all parameters across different urban locations. Krayenhoff et al. reflected spatial variation in d based on the building coverage ratio (BCR). Later on, Nazarian et al. and Lu et al. further extended spatial variation by introducing mathematical functions of urban characteristics to define α_2, d, and $C_\mu$dynamically. These approaches were primarily developed based on CFD simulations, which are mostly under idealized urban settings. One of the differences is the parameterization of turbulent viscosity (C_μ), which is shown in Figure 10. This parameterization involved multiple approaches, including both constant value and dynamic values. Even with the dynamic parameterizations proposed by Nazarian et al. and Lu et al., different trends are observed. Hence, the applicability of these parameterizations to real-world urban environments remains uncertain.

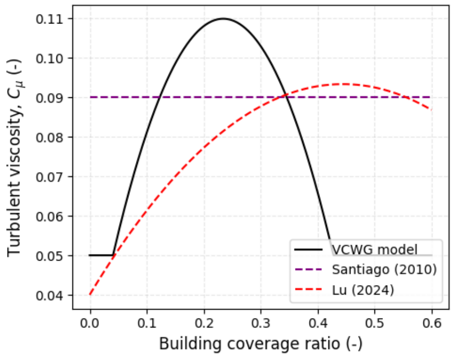

Figure 10. Comparison of turbulent viscosity parameterization in UCM models

Overall, our analysis results suggest that calibration of key parameters improves urban wind speed prediction, but additional refinement in some sub-models related to turbulence may further enhance the model performance. These include (1) dynamic adjustments to parameterization based on urban characteristics and (2) additional strategies to reduce the discrepancy of prediction due to missing urban characteristics representation in the model.

Conclusion

This study improved urban wind speed predictions by calibrating key parameters in the VCWG model. Calibration of key parameters reduced $RMSE_{all}$ from 1.81 m/s to 1.78 m/s. The calibration revealed trends, such as higher turbulent diffusion coefficients under unstable conditions and lower diffusion under stable conditions, suggesting that existing parameterizations underestimate urban turbulence. While this study focused on input parameter uncertainties, findings may suggest that model-form uncertainties—specifically in the parameterization of turbulence length scales derived from CFD studies—remain a significant limitation. Overall, this work provides evidence to correctly set up the physical parameters based on the calibration of UCMs under actual urban contexts.

However, this study has several limitations that may affect the applicability of the current UCM across different urban contexts. First, the calibration was conducted based on the measurement data from Seoul, Republic of Korea, which is classified as a humid continental climate. As a result, additional calibration across different climate zones may be necessary to generalize the findings. Second, while vegetation is included as an input variable in UCMs, the effects of urban vegetation were not explicitly considered due to the lack of detailed data.

https://doi.org/10.26868/25222708.2025.1459

Acknowledgement

This work was supported by the National Research Foundation of Korea(NRF) grant funded by the Korea government(MSIT) (No.2020R1A5A1018153). Gigih R. Setyantho is supported by the Hyundai Motor Chung Mong-Koo Global Scholarship (GSB-25-01864).

References

Aliabadi, A. A., Moradi, M., McLeod, R. M., Calder, D., & Dernovsek, R. (2021). How much building renewable energy is enough? The Vertical City Weather Generator (VCWG v1. 4.4). *Atmosphere*, *12*(7), 882.

Barlow, J. F. (2014). Progress in observing and modelling the urban boundary layer. *Urban Climate*, *10*, 216-240.

Blocken, B. (2015). Computational Fluid Dynamics for urban physics: Importance, scales, possibilities, limitations and ten tips and tricks towards accurate and reliable simulations. *Building and Environment*, *91*, 219-245.

Bueno, B., Norford, L., Hidalgo, J., & Pigeon, G. (2013). The urban weather generator. *Journal of Building Performance Simulation*, *6*(4), 269-281.

Chen, F., Kusaka, H., Bornstein, R., Ching, J., Grimmond, C. S. B., Grossman-Clarke, S., Loridan, T., Manning, K. W., Martilli, A., Miao, S., Sailor, D., Salamanca, F. P., Taha, H., Tewari, M., Wang, X., Wyszogrodzki, A. A., & Zhang, C. (2011). The integrated WRF/urban modelling system: development, evaluation, and applications to urban environmental problems. *International Journal of Climatology*, *31*(2), 273-288.

Herman, J., & Usher, W. (2017). SALib: An open-source Python library for sensitivity analysis. *Journal of Open Source Software*, *2*(9), 97.

Hong, T., & Heo, Y. (2021, 1-3 September). Spatio-temporal data analysis for development of microclimate prediction models. *Proceedings of the 17th IBPSA Building Simulation Conference*, Bruges (Belgium), 1-3 September 2021.

Hong, T., & Heo, Y. (2023). Exploring the impact of urban factors on land surface temperature and outdoor air temperature: A case study in Seoul, Korea. *Building and Environment*, *243*.

Krayenhoff, E. S., Santiago, J. L., Martilli, A., Christen, A., & Oke, T. R. (2015). Parametrization of Drag and Turbulence for Urban Neighbourhoods with Trees. *Boundary-Layer Meteorology*, *156*(2), 157-189.

Liu, Y., Zhou, Y., & Lu, J. (2020). Exploring the relationship between air pollution and meteorological conditions in China under environmental governance. *Sci Rep*, *10*(1), 14518.

Lu, J., Nazarian, N., Hart, M. A., Krayenhoff, E. S., & Martilli, A. (2024). A one-dimensional urban flow model with an eddy-diffusivity mass-flux (EDMF) scheme and refined turbulent transport (MLUCM v3. 0). *Geoscientific Model Development*, *17*(7), 2525-2545.

Mao, J., Yang, J. H., Afshari, A., & Norford, L. K. (2017). Global sensitivity analysis of an urban microclimate system under uncertainty: Design and case study. *Building and Environment*, *124*, 153-170.

Masson, V. (2000). A Physically-Based Scheme For The Urban Energy Budget In Atmospheric Models. *Boundary-Layer Meteorology*, *94*(3), 357-397.

McKay, M. D., Beckman, R. J., & Conover, W. J. (2000). A comparison of three methods for selecting values of input variables in the analysis of output from a computer code. *Technometrics*, *42*(1), 55-61.

Moradi, M. (2021). *The Vertical City Weather Generator* [Doctoral dissertation University of Guelph]. Zotero. Ontario, Canada.

Moradi, M., Dyer, B., Nazem, A., Nambiar, M. K., Nahian, M. R., Bueno, B., Mackey, C., Vasanthakumar, S., Nazarian, N., Krayenhoff, E. S., Norford, L. K., & Aliabadi, A. A. (2021). The Vertical City Weather Generator (VCWG v1.3.2). *Geoscientific Model Development*, *14*(2), 961-984.

Morris, C. J. G., & Simmonds, I. (2001). Quantification of the Influences of Wind and Cloud on the Nocturnal Urban Heat Island of a Large City. *Journal of Applied Meteorology and Climatology*, *40*(2), 169–182.

Nazarian, N., Krayenhoff, E. S., & Martilli, A. (2020). A one-dimensional model of turbulent flow through “urban” canopies (MLUCM v2.0): updates based on large-eddy simulation. *Geoscientific Model Development*, *13*(3), 937-953.

Oke, T. R. (2002). *Boundary layer climates*. Routledge.

Ricciardelli, F., & Polimeno, S. (2006). Some characteristics of the wind flow in the lower Urban Boundary Layer. *Journal of Wind Engineering and Industrial Aerodynamics*, *94*(11), 815-832.

Rotach, M., Vogt, R., Bernhofer, C., Batchvarova, E., Christen, A., Clappier, A., Feddersen, B., Gryning, S.-E., Martucci, G., & Mayer, H. (2005). BUBBLE–an urban boundary layer meteorology project. *Theoretical and Applied Climatology*, *81*(3), 231-261.

Santiago, J. L., & Martilli, A. (2010). A Dynamic Urban Canopy Parameterization for Mesoscale Models Based on Computational Fluid Dynamics Reynolds-Averaged Navier–Stokes Microscale Simulations. *Boundary-Layer Meteorology*, *137*(3), 417-439.

Setyantho, G. R., Yuan, C., & Heo, Y. (2024). Evaluation of multi-layer urban canopy model (MLUCM) for urban microclimate predictions at different urban contexts. *Urban Climate*, *55*.

United Nations Department of Economic and Social Affairs. (2019). *Urban and rural population growth and world urbanization prospects* (World urbanization prospects: The 2018 revision, Issue.

Wang, W., Ng, E., Yuan, C., & Raasch, S. (2017). Large-eddy simulations of ventilation for thermal comfort — A parametric study of generic urban configurations with perpendicular approaching winds. *Urban Climate*, *20*, 202-227.

https://doi.org/10.26868/25222708.2025.1459